Year 7 Mathematics

Unit 2: Number Skills

2.1 Rules of Divisibility

Rules of Divisibility

STARTER · 5 MIN

Date: _____ **Title: Rules of Divisibility**

LEARNING OBJECTIVES

- To be able to use rules for divisibility by 2, 3, 4, 5, 9 and 10

STARTER *Answer in your book. No calculators.*

1. Is 406 divisible by 2?
2. Is 95 divisible by 5?
3. Is 2 000 divisible by 10?
4. Is 77 divisible by 2?

Starter — Answers

STARTER

Tick or correct in green pen

1. **Yes: it is even**
2. **Yes: it ends in 5**
3. **Yes: it ends in 0**
4. **No: it is odd**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can state the divisibility rules for 2, 3, 4, 5, 9 and 10	☐
SILVER	I can use the rules to test if a number is divisible by these divisors	☐
GOLD	I can explain why these rules work using examples	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Rules of Divisibility

INTRODUCTION · 3 MIN

A number is divisible by another if it divides exactly, with no remainder.

2, 5 and 10

Look at the last digit.

2: even 5: 0 or 5 10: 0

3 and 9

Add the digits.

Digit sum divisible by 3 (or by 9)

4

Look at the last two digits.

1 044: 44 ÷ 4 = 11, so yes

Divisible

Divides exactly: no remainder.

Divisible by 5 = a multiple of 5

Examples 1 and 2 — Dividing by 2, 5 and 10

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint 2: even 5: ends 0 or 5 10: ends in 0

Example 1

Which of these numbers are divisible by 2?

8, 77, 1 000, 406, 189

Example 2

Which of these numbers are divisible by 5? By 10?

58, 2 000, 95, 1 949

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Divisible by 2: the number is even

8, 1 000 and 406 end in 0, 2, 4, 6 or 8

77 and 189 are odd: not divisible

Tip: only the last digit matters.

Example 2

5: ends in 0 or 5. 10: ends in 0

By 5 = 2 000 and 95

By 10 = 2 000

Tip: 58 and 1 949 do not end in 0 or 5.

Task 1

PRACTICE · 9 MIN

BRONZE *Use the rules for 2, 5 and 10*

Hint 2: even 5: ends 0 or 5 10: ends in 0

1. Is 74 divisible by 2?

2. Is 135 divisible by 5?

3. Is 3 470 divisible by 10?

4. Which are divisible by 2?
   15, 22, 40, 63

5. Which are divisible by 5?
   25, 52, 70, 101

★ Which are divisible by 2, 5 and 10? 30, 45, 100, 108

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **Yes: it is even**
2. **Yes: it ends in 5**
3. **Yes: it ends in 0**
4. **22 and 40**
5. **25 and 70**

★ **30 and 100**

Examples 3 and 4 — Dividing by 3, 4 and 9

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint 3 and 9: add the digits 4: last two digits

Example 3

Are these numbers divisible by 3? By 9?

189, 2 058, 325

Example 4

Are these numbers divisible by 4?

104, 1 044, 320

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Add the digits

189 = 1 + 8 + 9 = 18

18: divisible by 3 and by 9

2 058 = 2 + 0 + 5 + 8 = 15

15: divisible by 3, not by 9

325 = 3 + 2 + 5 = 10

10: not divisible by 3 or 9

Tip: a digit sum of 9, 18, 27 ... means ÷ 9.

Example 4

Look at the last two digits

104 = 04 ÷ 4 = 1

1 044 = 44 ÷ 4 = 11

320 = 20 ÷ 4 = 5

All three are divisible by 4.

Tip: 100 is divisible by 4, so ignore the hundreds.

Task 2

PRACTICE · 9 MIN

SILVER *Use the rules for 3, 4 and 9*

Hint 3 and 9: add the digits 4: last two digits

1. Is 252 divisible by 3?

2. Is 418 divisible by 3?

3. Is 738 divisible by 9?

4. Is 316 divisible by 4?

5. Is 1 030 divisible by 4?

★ Which are divisible by 3?
24, 35, 72, 104, 135

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **Digit sum = 9**
   Yes

2. **Digit sum = 13**
   No

3. **Digit sum = 18**
   Yes

4. **16 ÷ 4 = 4**
   Yes

5. **30 ÷ 4 has a remainder**
   No

★ **24, 72 and 135**

Examples 5 and 6 — Reasoning with Divisibility

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Counterexample: one case that fails
Divisible by 2 and 5 = by 10

Example 5

(2 marks)

A number is 3 2 4 □.
The last digit is missing.

Which digits make it divisible by both 2 and 5? Explain.

Example 6

(2 marks)

A number divisible by 9 is also divisible by 3.

Is a number divisible by 3 always divisible by 9?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Divisible by 2 and 5: ends in 0

By 2: last digit 0, 2, 4, 6 or 8

By 5: last digit 0 or 5

Only 0 is in both lists.

Missing digit = 0

Tip: divisible by 2 and 5 means divisible by 10.

Example 6

No: give a counterexample

12: digit sum 3, so 12 ÷ 3 = 4

but 12 ÷ 9 has a remainder.

9 = 3 × 3, so every multiple of 9

is a multiple of 3, not the other way.

Tip: one counterexample proves it false.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: explain your answers

Hint Counterexample: one case that fails
Divisible by 2 and 5 = by 10

1. Find the missing digit:
   5 □ 2 is divisible by 9

2. Which digits make 7 □
   divisible by 3?

3. Is 2 345 divisible by 5?
   By 3? Explain.

4. Find a number between 40
   and 50 divisible by 3 and 4.

5. A number is divisible by 2
   and 3. Is it divisible by 6?

★ Find the smallest number
divisible by 4, 5 and 9.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $5 + \square + 2 = 9$

$\square = 2$

2 72, 75, 78

$\square = 2, 5$ or 8

3 By 5: yes, ends in 5

By 3: no, digit sum 14

4 48

5 Yes: $6 = 2 \times 3$

e.g. $18 \div 6 = 3$

★ $4 \times 5 \times 9 = 180$

(4, 5 and 9 share no factors)

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Is 345 divisible by 5?

SILVER Is 456 divisible by 3? Show the digit sum.

GOLD Divisible by 9 → divisible by 3, but not always back. Why?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **Yes: it ends in 5**

SILVER **4 + 5 + 6 = 15**

Yes

GOLD **9 = 3 × 3. 12 ÷ 3 = 4, but 12 ÷ 9 has a remainder.**

Exam Practice — Divisibility by 2, 5 and 10

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint 2: even 5: ends 0 or 5 10: ends in 0

Question 1

(2 marks)

Here is a list of numbers.

48 55 70 93 120

(a) Which are divisible by 2? (1)

.....

(b) Which are divisible by 10? (1)

.....

Question 2

(2 marks)

Here is a list of numbers.

70 85 92 100 115

(a) Which are divisible by 5? (1)

.....

(b) Which are divisible by 2 and 5? (1)

.....

Exam Practice — Divisibility by 3, 4 and 9

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint 3 and 9: add the digits 4: last two digits

Question 3

(2 marks)

Write down all the numbers in this list that are divisible by 3.

27 41 63 100 114

.....

Question 4

(3 marks)

5 742 and 1 128

(a) Is 5 742 divisible by 9? Show working. (2)

.....

(b) Is 1 128 divisible by 4? (1)

.....

Exam Practice — Reasoning with Divisibility

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Counterexample: one case that fails
Divisible by 2 and 5 = by 10

Question 5

(2 marks)

A number is 5 7 1 □.
The final digit is missing.

Write down every final digit that makes the number divisible by 2 and 5.
Explain your answer.

.....

Question 6

(3 marks)

Ali says: every number divisible by 3 is also divisible by 9.

(a) Give a counterexample. (1)

.....

(b) Find the smallest number divisible by 2, 3 and 5. (2)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 48, 70, 120 — B1

(b) 70, 120 — B1

Tip: even numbers end in 0, 2, 4, 6 or 8.

Question 2

(a) 70, 85, 100, 115 — B1

(b) 70, 100 — B1

Tip: by 2 and by 5 means it ends in 0.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

27, 63, 114 — B2

B1 for two correct, none wrong

Digit sums: 9, 5, 9, 1, 6

Tip: add the digits of each number.

Question 4

(a) 5 + 7 + 4 + 2 = 18 — M1

Yes: 18 is divisible by 9 — A1

(b) Yes: 28 ÷ 4 = 7 — B1

Tip: for 4, test the last two digits.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

0 B1

By 2: even. By 5: ends 0 or 5. B1

Only 0 does both.

Tip: list the digits for each rule.

Question 6

(a) e.g. 6: not divisible by 9 B1

(b) 2 × 3 × 5 = 30 M1 A1

Tip: check 30 ÷ 2, 30 ÷ 3 and 30 ÷ 5.